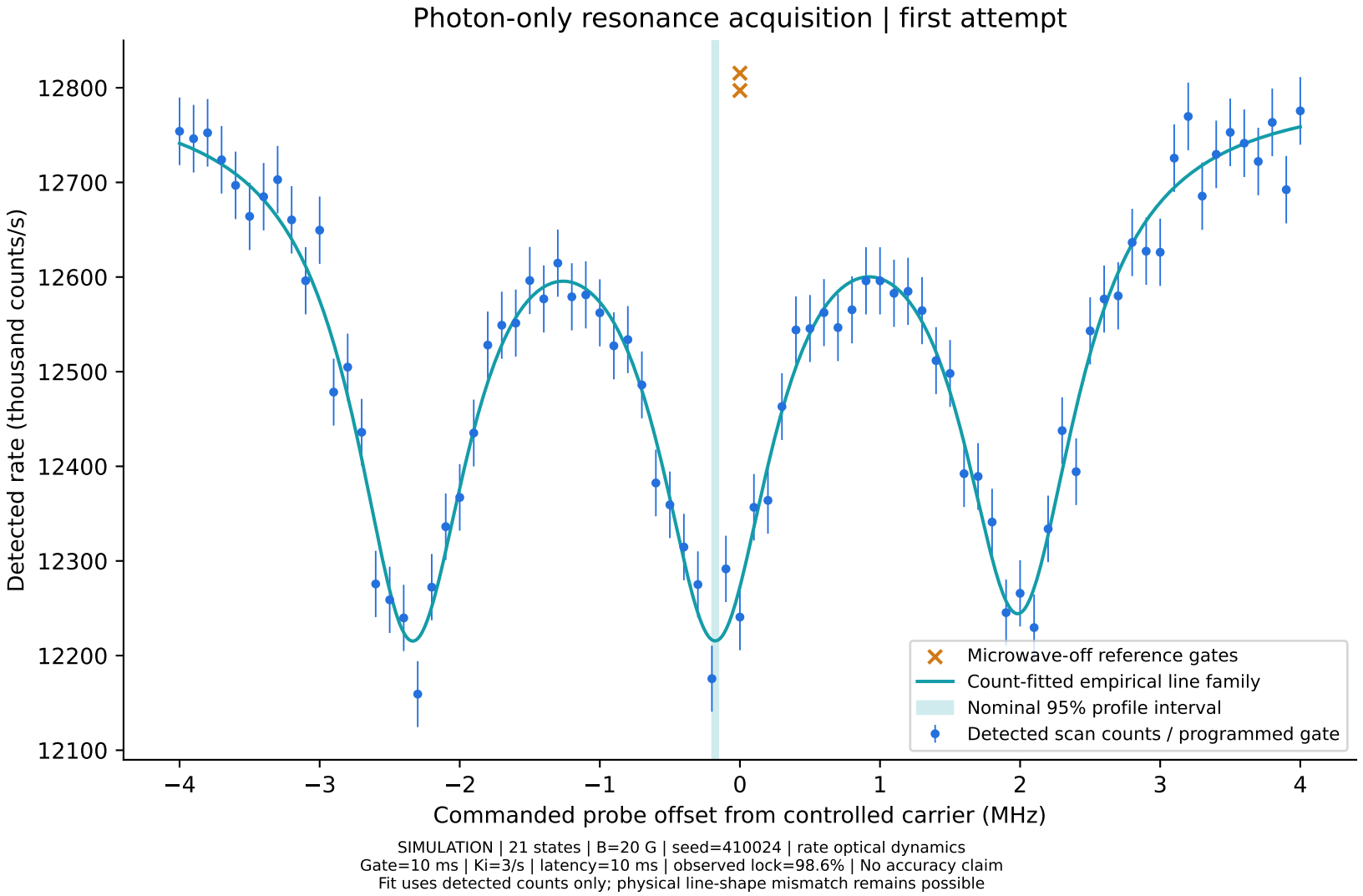
Photon-only resonance acquisition | first attempt
Detected rate (thousand counts/s)
12800
12700
12600
12500
12400
12300
12200
12100
−4
−3
−2
−1
0
1
2
3
4
Commanded probe offset from controlled carrier (MHz)
Microwave-off reference gates
Count-fitted empirical line family
Nominal 95% profile interval
Detected scan counts / programmed gate
SIMULATION | 21 states | B=20 G | seed=410024 | rate optical dynamics
Gate=10 ms | Ki=3/s | latency=10 ms | observed lock=98.6% | No accuracy claim
Fit uses detected counts only; physical line-shape mismatch remains possible